EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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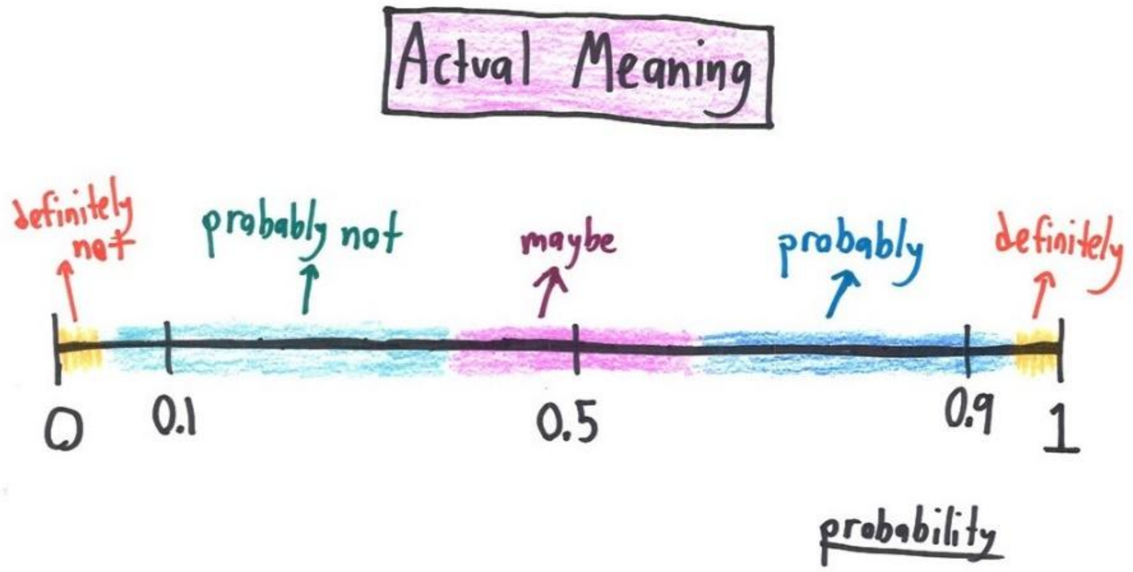
Week 3 - Lecture No 05 – 3rd September 2025



Announcements!

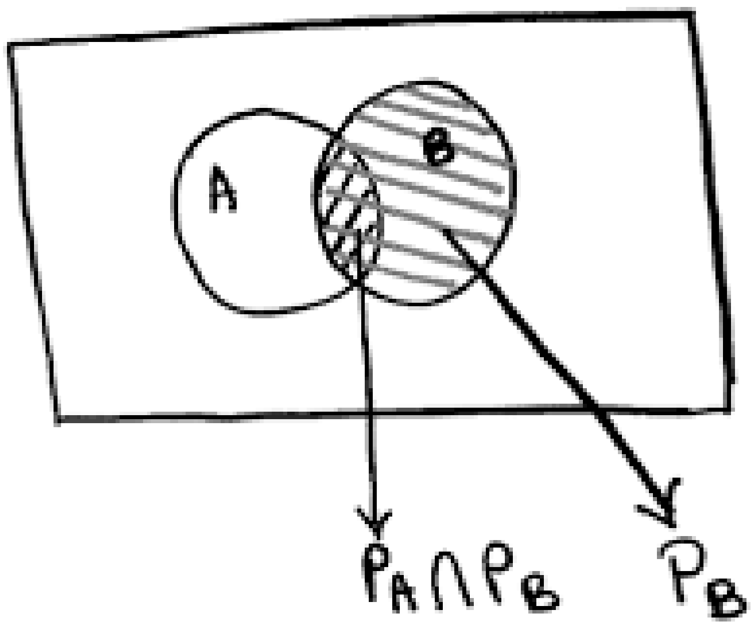
1. Lectures up to No 04 posted

Agenda for today

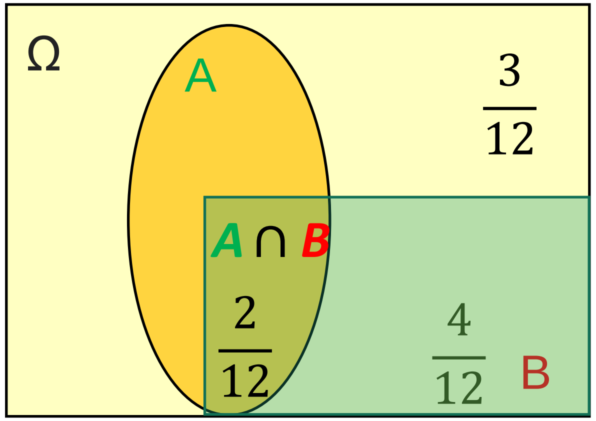
1. Quiz No 01
2. Unit 2 - Conditional Probability and Bayes' Rule

Unit 2: Conditional Probability and Bayes' Rule

- Conditional Probability
- Three important Tools
  - Multiplication rule
  - Total probability theorem
  - Bayes' rule (→ Inference)



Definition of conditional probability



- $\mathbf{P}(A|B)$ = "probability of A, given that B occurred"

$$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} = \frac{2/12}{6/12} = \frac{1}{3}$$

defined only when $\mathbf{P}(B) > 0$

Example: Models based on conditional probabilities.

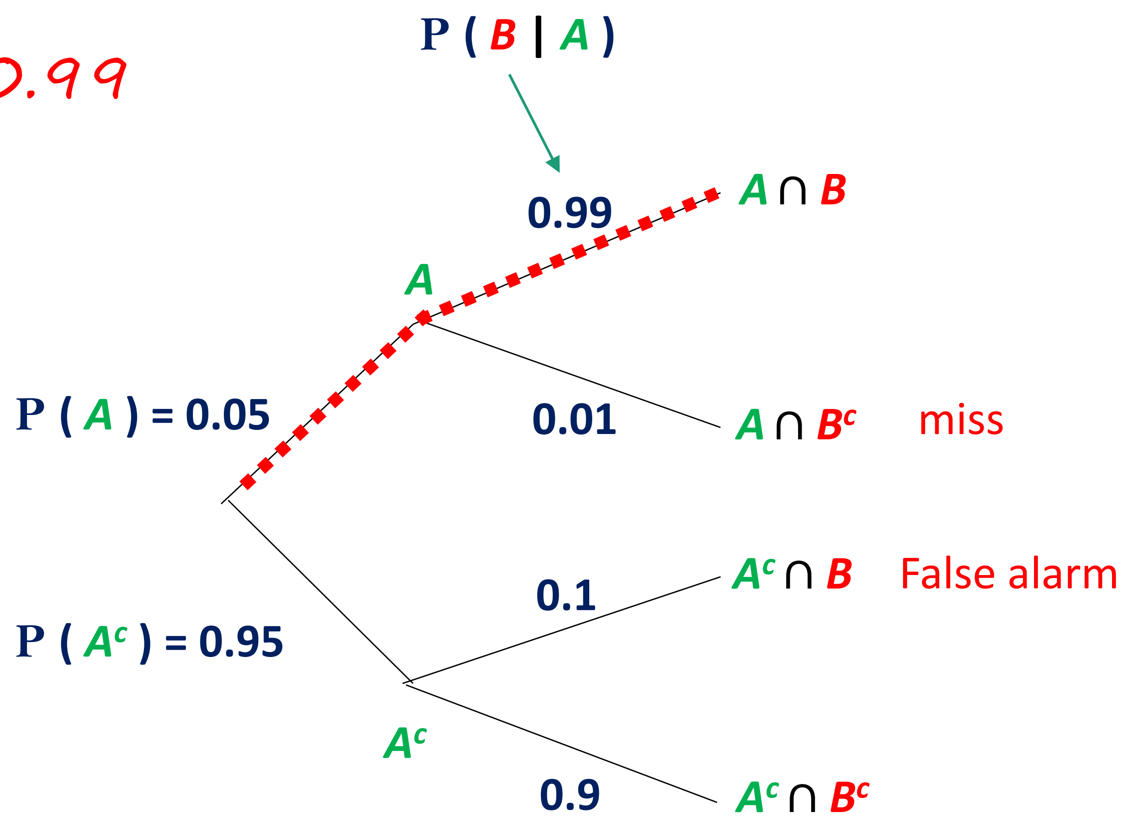
- Event A: Airplane is flying above.
- Event B: Something registers on radar screen

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

❖ **Problem - Find The probability an aircraft is flying given radar registers something on radar screen – P(A|B) ?**

- $P(A \cap B) = P(A) * P(B|A) = 0.05*0.99$
- $P(B) = 0.05*0.99 + 0.95*0.1$
  $= 0.1445$
- $P(A|B) = \dfrac{0.05*0.99}{0.1445} = 0.34$

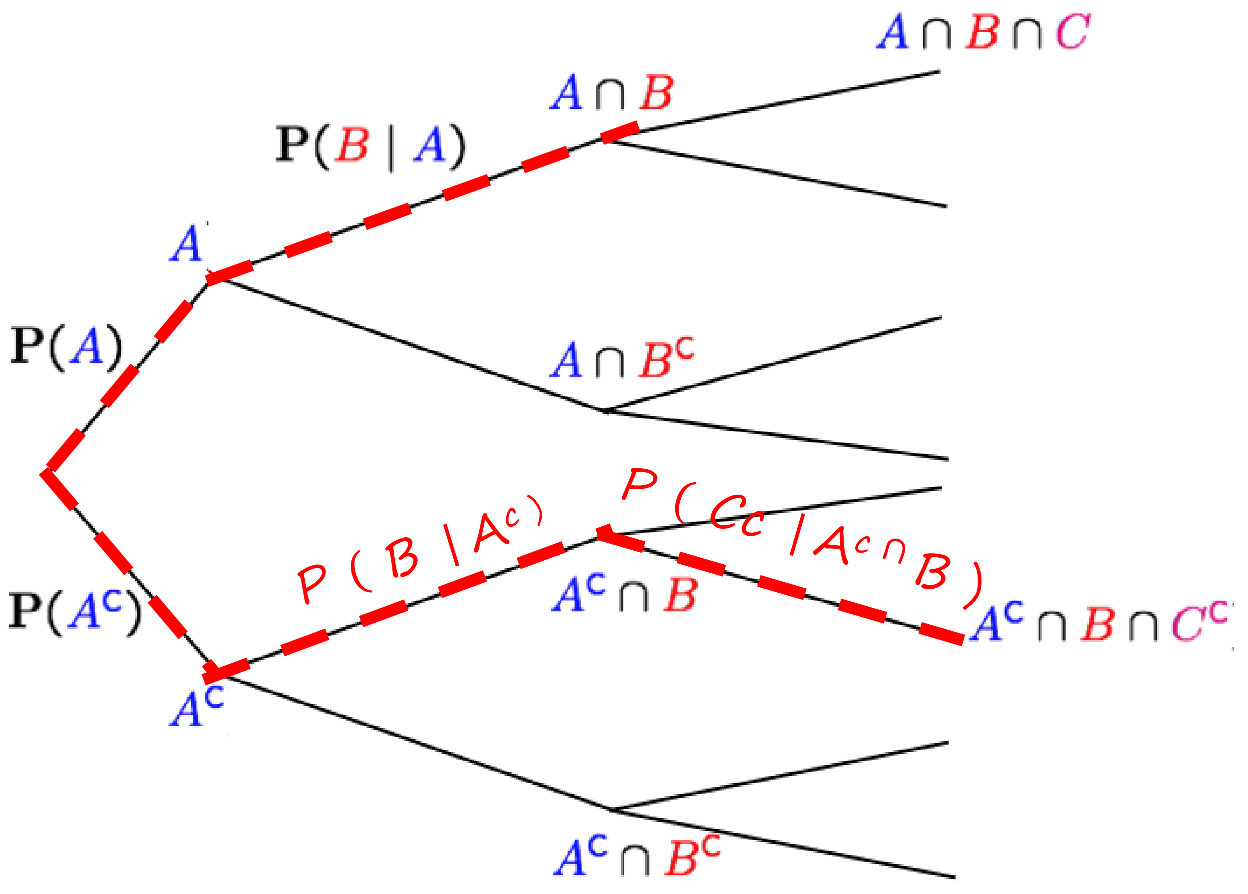


Multiplication Rule

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A \mid B)\,P(B)$$

$$= P(B \mid A)\,P(A)$$



$$P(A^c \cap B \cap C^c) =$$

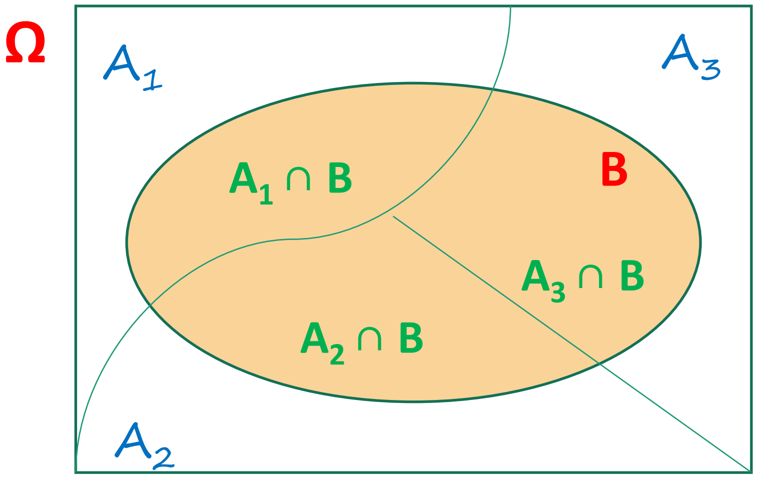
$$= P(A^C \cap B)\,P(C^C \mid A^C \cap B)$$

$$= P(A^C)\,P(B \mid A^C)\,P(C^C \mid A^C \cap B)$$

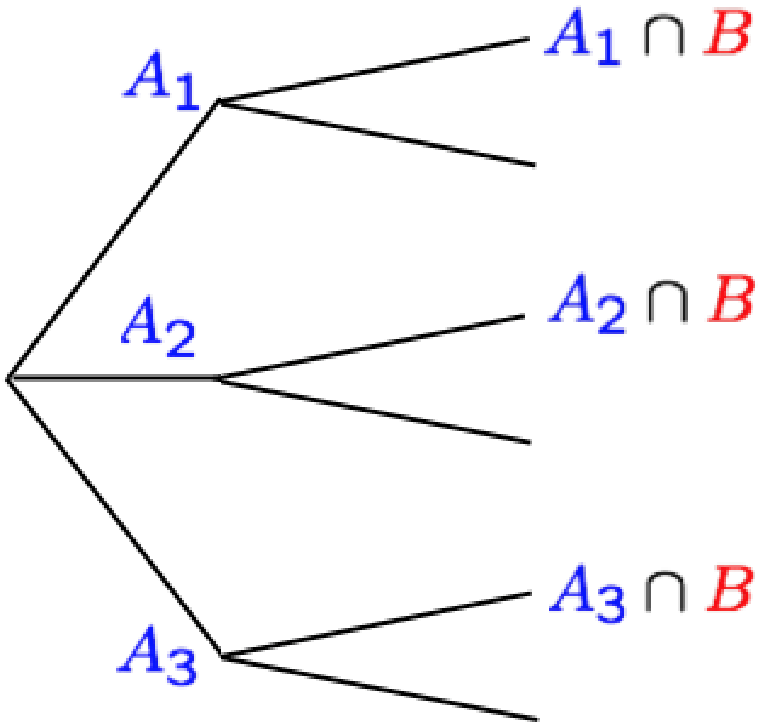
$$P(A_1 \cap A_2 \cap A_3 \cap \ldots \cap A_4)$$

$$= P(A_1) \prod_{i=2}^{n} P(A_i \mid (A_1 \cap A_2 \cap \ldots \cap A_{i-1}))$$

Total Probability Theorem



- Partition of sample space in A_1, A_2, A_3
- Have $\mathbf{P}(A_i)$, for all *i*.
- Have $\mathbf{P}(B \mid A_i)$, for all *i*.



➡ $\mathbf{P}(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$

➡ $= P(A_1)\,P\left(B \middle| A_1\right) + \ldots + \ldots$

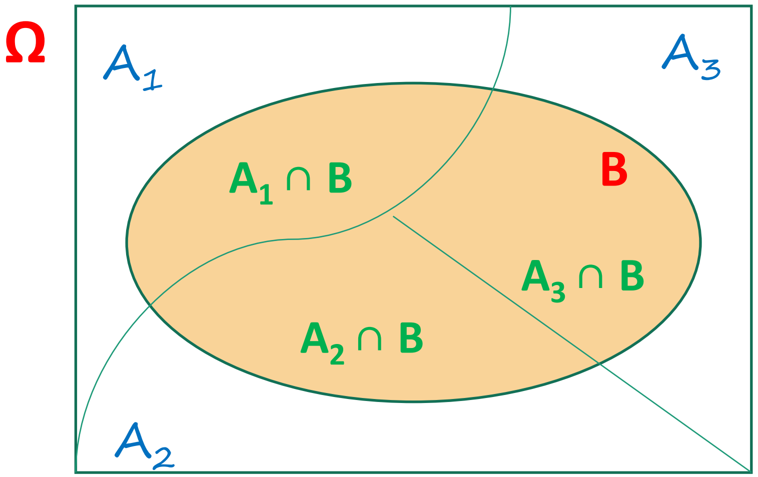
➡ $\sum_j P(A_j) = 1$

weights

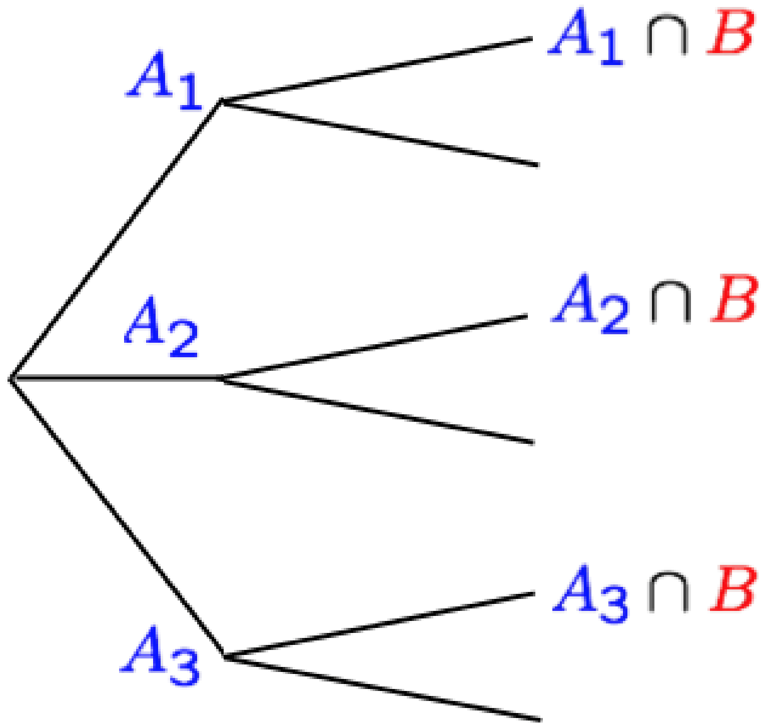
$$\boxed{\mathbf{P}(B) = \textstyle\sum_j \mathbf{P}(Aj)\,\mathbf{P}(B \mid Aj)}$$

weighted average of $P(B \mid A_i)$

Bayes Rule



- Partition of sample space in A_1, A_2, A_3
- Have $\mathbf{P}(A_i)$, for all *i*. initial "beliefs",
- Have $\mathbf{P}(B \mid A_i)$, for all *i*.

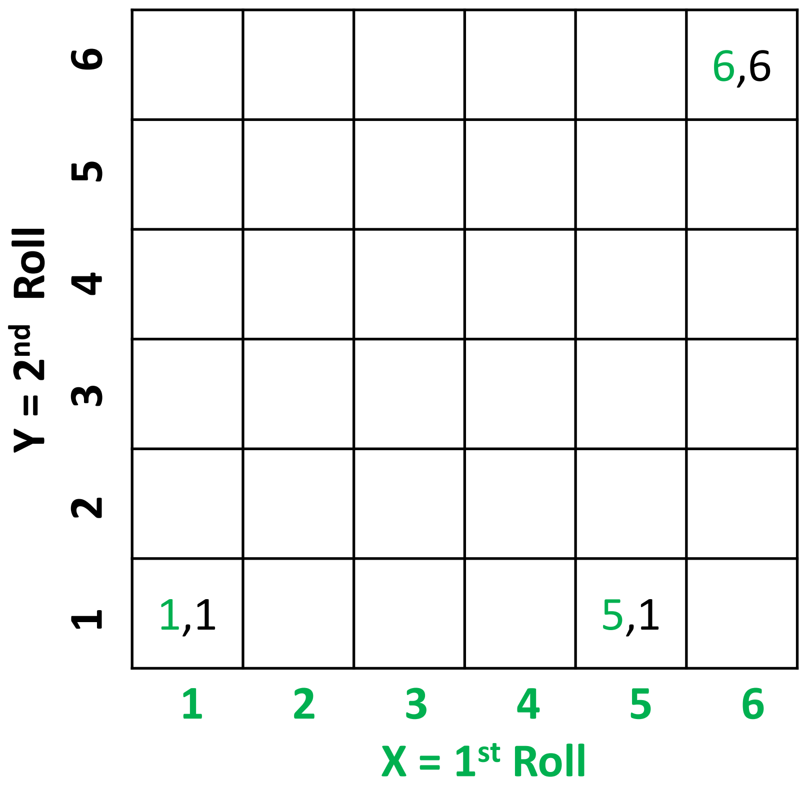


$$\mathbf{P}(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)}$$

Revised "beliefs", given that B occurred

$$\boxed{\mathbf{P}(A_i \mid B) = \frac{\mathbf{P}(A_i)\,\mathbf{P}(B \mid A_i)}{\sum_j \mathbf{P}(A_j)\,\mathbf{P}(B \mid A_j)}}$$

Example: two rolls of a fair 6-sided die – conditioning Example



D = two rolls resulting in doubles – (1,1), (2,2), ...

S = at least one of the rolls is a six

K = the rolls are of different numbers

(a) P(D) = 6/36

(b) $P(D|\text{Sum} \leq 4) = \dfrac{2/36}{6/36} = 2/6 = 1/3$

(c) P(S) = 11/36

(d) P(S|K) = 10/30 = 1/3

Example: Conditioning Example

We toss a biased coin twice with,

Event A = 1st toss is a head, Event B = 2nd toss is a head

Show that $P(A \cap B \mid A) \geq P(A \cap B \mid (A \cup B))$

Bayes' Rule & inference

- Thomas Bayes, Presbyterian minister (c. 1701-1765)
- "Bayes' Theorem" published posthumously
- Systematic approach for incorporating new evidence

Bayesian Inference

- Initial beliefs $P(A_i)$ on all causes of an observed event B.
- Model of the world under each A_i: $P(B \mid A_i)$

$$A_i \xrightarrow[P(B \mid A_i)]{\textbf{Model}} B$$

- Draw conclusions about causes

$$B \xrightarrow[P(A_i \mid B)]{\textbf{inference}} A_i$$

1) $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

2) $P(B|A) = \dfrac{P(A \cap B)}{P(A)}$

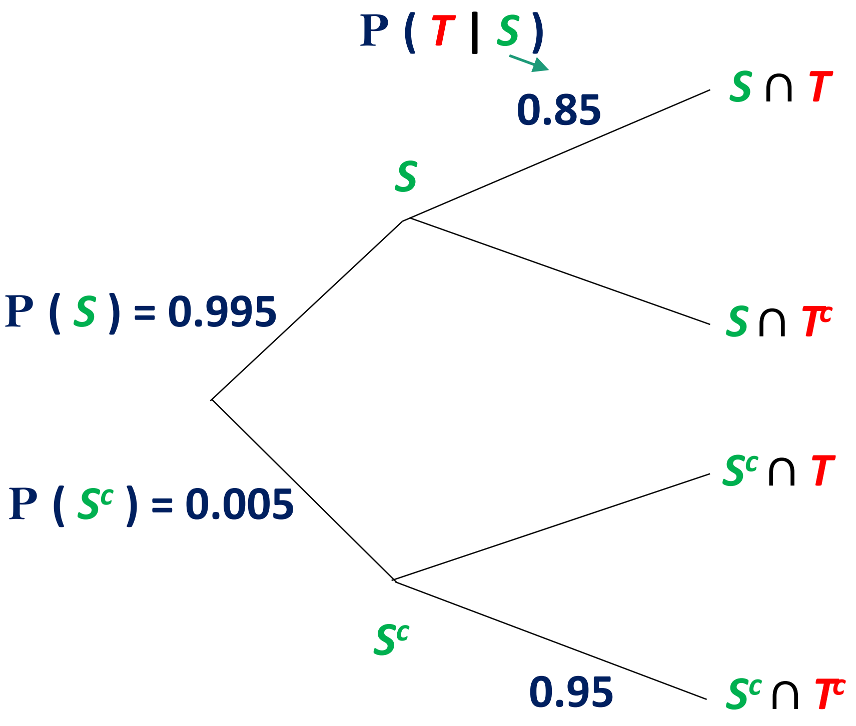
3) $P(B|A) = \dfrac{P(A|B) \, . \, P(B)}{P(A)}$

Example: Baye's Rule

A new test (T) has been developed to determine whether a given student is overstressed (S). This test is 95% accurate if the student is not overstressed, but only 85% accurate if the student is in fact overstressed. It is known that 99.5% of all students are overstressed. **Given that a particular student tests negative for overstress, what is the probability that the test results are correct, and that this student is not overstressed?**

S = Student is over-stressed, S^c = Student is NOT over-stressed

T = Test positive, T^c = Test negative



$$\textbf{Find } P(S^c | T^c) = \frac{P(T^c|S^c) \times P(S^c)}{P(T^c)}$$

$$= \frac{0.95 \times 0.005}{0.95 \times 0.005 + 0.15 \times 0.995}$$

$$= 0.0308$$

1) $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

2) $P(B|A) = \dfrac{P(A \cap B)}{P(A)}$

3) $P(B|A) = \dfrac{P(A|B) . P(B)}{P(A)}$